

Chapter 8

Intricacies of the Elaboration Likelihood Model

Introduction

We have now presented the major postulates of the Elaboration Likelihood Model and the evidence for these postulates. In reviewing the evidence for the ELM we have focused deliberately on variables and instances that were straightforward and relatively unambiguous in interpretation. Although it would be nice if we could have provided an exhaustive list of variables that serve as peripheral cues and variables that affect message processing in either a relatively objective or a biased manner, we have already seen that this is not possible. For example, we have argued that the effects on information processing of some variables may shift from relatively objective to relatively biased as the variable reaches very high levels. Thus, although increasing personal relevance and message repetition may generally enhance subjects' motivation and/or ability to see the merits of strong arguments and the flaws in weak ones, we have suggested that when personal relevance or message repetition reach very high levels, the initially objective processing may become biased as the person becomes motivated to reject the advocacy (Cacioppo & Petty, 1979b; Petty & Cacioppo, 1979b). In short, some variables have multiple effects on information processing. In addition, we have seen that some variables may affect information processing under certain conditions, but serve as peripheral cues in other contexts. For example, we reviewed evidence in Chapter 6 that a manipulation of the number of arguments in a message could serve as a peripheral cue when the personal relevance of the message was low, but that increasing the number of arguments in a message could increase the amount of information processing activity when the personal relevance of the message was high (Petty & Cacioppo, 1984a). In this chapter we will comment on these and other intricacies of the ELM.

Variables with Multiple Effects on Elaboration

In most of the research that we have discussed so far, we have examined the isolated effects of different source, message, recipient, and channel factors on information processing. However, in most natural persuasion situations, many variables combine to create the overall persuasion context. For example, consider a high need for cognition person who is part of a jury whose members share responsibility for evaluating an expert witness who presents weak arguments in a corporate tax case in a courtroom with noisy distractions. All of the many variables present in this situation must be considered jointly to determine the probable persuasive impact of the testimony. Normally, sharing cognitive responsibility with a group reduces information processing activity (Petty, Cacioppo, & Harkins, 1983), but our message recipient dispositionally tends to like to think (Cacioppo, Petty, & Morris, 1983) and is therefore less susceptible to motivation loss in groups when the task is an interesting cognitive one (Petty, Cacioppo, & Kasmer, 1985). Therefore, *motivation* to process the message is likely high despite the group responsibility. However, due to a lack of prior knowledge about corporate taxes and the distractions inherent in the situation, our message recipient may have little *ability* to process the message arguments (Petty, Wells, & Brock, 1976; Wood, Kallgren, & Priesler, 1985). Thus, the perceived expertise of the witness may serve as a potent influence cue (Kiesler & Mathog, 1968).¹

Our example relies mostly on the idea that each of the features of the persuasion situation (e.g., distraction, group responsibility) can be considered separately and independently regardless of the levels of the other variables with which it is combined. If so, one can roughly add (subtract) the effects of each variable to determine the overall elaboration likelihood. Although this is sometimes possible, as we discuss next, it is also possible for one variable to have very different effects on information processing depending on the level of other variables.

Variables that Enhance and Reduce Message Processing

The Effect of Rhetorical Questions

Some variables may increase information processing at one level of another factor, but may actually decrease processing at a different level of that factor.

¹Just as any one persuasion context may contain many variables that affect the elaboration likelihood, any one persuasion context may contain multiple peripheral cues. We assume that the less motivated a person is to scrutinize an advocacy, the fewer the cues that will have an impact on attitudes. When motivation is very low, the most salient cue will likely have the largest impact on attitudes. On the other hand, if motivation is high, but ability to process issue-relevant arguments is low, people may scrutinize the message and persuasion context very carefully and consider all relevant cues in order to evaluate the advocacy.

For example, we hypothesized that the use of rhetorical questions could operate in this manner. Zillmann (1972) reported the first experimental investigation of rhetorical questions. In his study, subjects heard a defense attorney's closing arguments in a second degree murder case. The subjects either heard the summary presented completely in statement form or with 10 argument-condensing persuasive statements transformed to rhetorical form (e.g., "Johnny was a peaceful boy," to "Johnny was a peaceful boy, wasn't he?"). Subjects hearing the rhetorical version were more favorable to the defense than those hearing the statement version. Zillmann proposed that rhetorical questions tended to be used with strong arguments and that through socialization, rhetorical questions would come to "mark relatively powerful arguments" for most people (i.e., rhetorical questions served as a positive peripheral cue). We suggested, on the other hand, that given people had the ability to process the issue-relevant arguments in the message, that the use of rhetorical questions would motivate thinking about the argument associated with the rhetorical question. If people attempted to covertly respond to the rhetorical questions in a message, then the use of rhetorical questions would likely enhance agreement only if the message arguments were strong. If the arguments were weak, people would likely disagree with the rhetorical questions and show less agreement than if the message contained no rhetorical questions. According to the positive cue hypothesis, rhetorical questions should enhance persuasion for both strong and weak messages.

If the elaboration view is correct, however, another point to consider is that the use of rhetorical questions should be most effective in enhancing thinking when recipients are not naturally devoting much effort to scrutinizing the message. If the message was naturally eliciting a great deal of thought, it would be unlikely that the use of rhetorical questions could enhance elaboration further. Under these conditions, rhetorical questions might have no further effect on elaboration and persuasion, or the use of rhetorical questions might actually interfere with subjects' thought processes. Based on pilot work, we favored the latter possibility. Specifically, in a preliminary study, subjects were exposed to a personally relevant audiotaped message that either did or did not contain rhetorical questions. When the message contained rhetorical questions, subjects reported that although they were trying very hard to think about the message, the questions asked by the speaker disrupted their ongoing chain of thoughts. This distraction effect of rhetorical questions is likely confined to situations in which the presentation rate of the message is controlled externally rather than by the subject. If the presentation rate was controlled by the subject (e.g., a print message), then the disruption effect should be eliminated. In any case, to test our view that rhetorical questions might enhance processing when the motivation to elaborate was low, but inhibit ongoing processing when motivation was already high (as least for audio messages), we conducted a study in which we varied the use of rhetorical questions and argument quality along with a variable known to affect subjects' motivation to process issue-relevant arguments—the personal relevance of the message (Petty, Cacioppo, & Heesacker, 1981).

Method. A total of 160 students participated in a 2 (Message form: statement or rhetorical) $\times$ 2 (Personal relevance: low or high) $\times$ 2 (Argument quality: strong or weak) between-subjects factorial design. Subjects believed that their task was to evaluate radio editorials and they heard one of four versions of an audiotaped editorial advocating that seniors be required to pass a comprehensive exam in their major as a requirement for graduation. Subjects in the high relevance conditions were led to believe that the president of their university had recommended that the exam proposal begin at their university in the next year. Subjects in the low relevance conditions were led to believe that the president of a distant university had recommended that the exam proposal be initiated at his institution in 10 years. Subjects heard either a strong or a weak editorial containing eight major arguments in favor of instituting comprehensive exams. In the regular editorials, each of the major arguments ended with a summary sentence in the form of a declarative statement (e.g., Thus, instituting a comprehensive exam would be an aid to those who seek admission to graduate and professional schools). In the rhetorical versions of the message, six of the eight argument summarizing statements were transformed into rhetorical questions (e.g., Wouldn't instituting a comprehensive exam be an aid to those who seek admission to graduate and professional schools?). Following exposure to the appropriate editorial, subjects completed attitude measures, were given 2.5 min to list their thoughts, and responded to various ancillary measures.

Results. Analyses of the listed thoughts revealed that the strong and weak messages differed in the profile of thoughts elicited as expected. Of greater interest, however, was a three-way interaction on the measure of attitude toward senior comprehensive exams. This interaction revealed that when personal relevance was low, attitudes in response to the strong and weak messages were more extreme with the rhetorical than the statement version of the advocacy (see top left panel of Figure 8-1), but when personal relevance was high, attitudes in response to the strong and weak messages were more extreme with the statement than the rhetorical version of the communication (see top right panel of Figure 8-1). In addition, subjects were asked how distracted they were from thinking about the message. A significant Message form $\times$ Personal relevance interaction revealed that low relevance subjects felt less distracted with the rhetorical than the statement advocacy, but high relevance subjects felt more distracted with the rhetorical than the statement message. This pattern of data is consistent with the view that the use of rhetorical questions increased elaboration when subjects were not naturally motivated to process the issue-relevant arguments, but inhibited elaboration of the arguments when processing was already high. In sum, it appears that summarizing arguments as rhetorical questions can either increase or decrease elaboration of audio communications depending upon the normal level of processing elicited by the message.

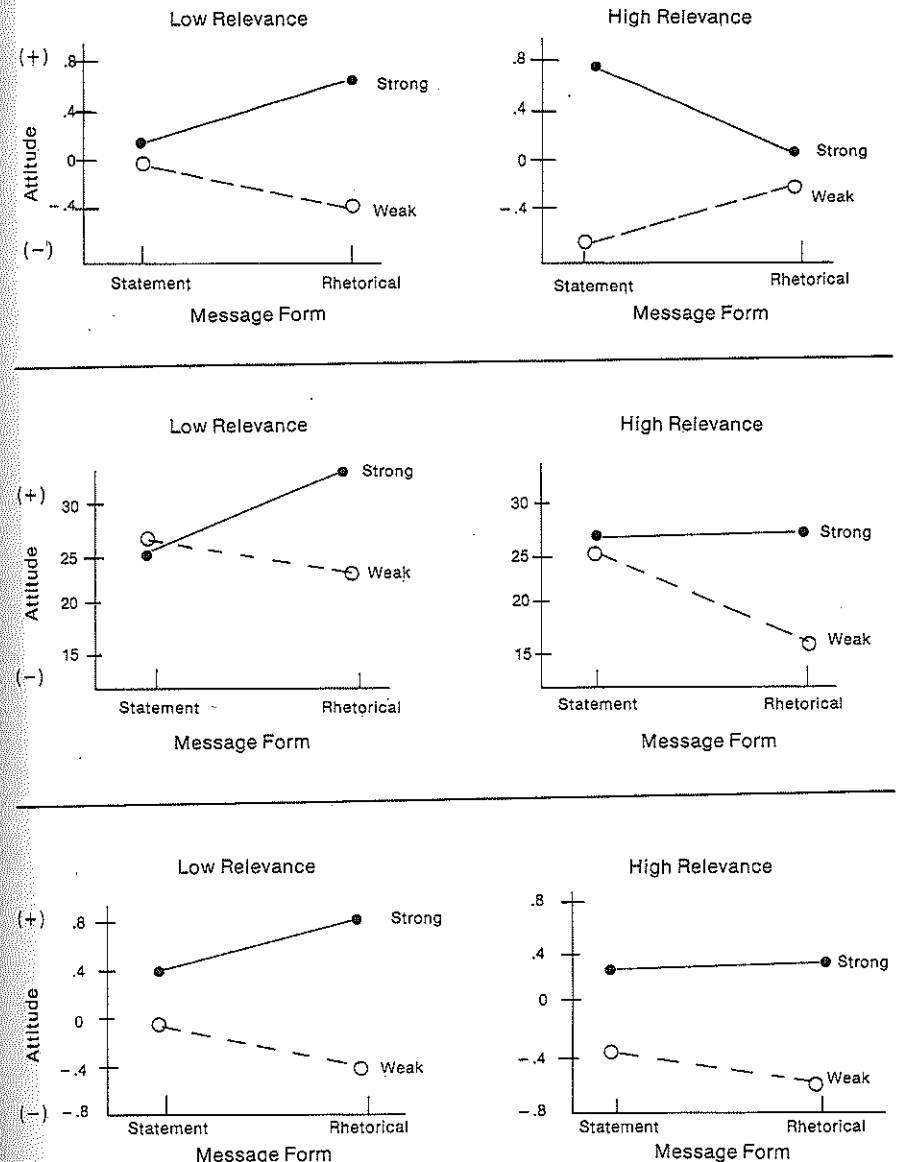


Figure 8-1. Postmessage attitudes as a function of personal relevance, argument quality, and the use of rhetorical questions. Figure shows that across three replications, the use of rhetorical questions enhanced message processing under conditions of low personal relevance. The effects of rhetoricals under high relevance were less clear (see text for discussion). (Top panel—data from Petty, Cacioppo, & Heesacker, 1981. Middle panel—data from Burnkrant & Howard, 1984. Bottom panel—data from Swasy & Munch, 1985.)

Additional Research on Rhetorical Questions

Recently, Burnkrant and Howard (1984) conducted a replication of our rhetorical study making a few key changes. Subjects were presented with the strong or weak version of our senior comprehensive exam message that was made either high or low in personal relevance. Instead of hearing the message over headphones, however, subjects were presented with the message as a written communication. In addition, instead of summarizing each message argument as a rhetorical question after the argument was presented, all rhetorical questions *preceded* the presentation of the arguments. Burnkrant and Howard argued that these changes should eliminate the distraction effect that we observed under high involvement. First, as we noted earlier, presenting the message in print rather than orally gives subjects time to stop the message to consider the arguments fully. Thus, the rhetorical questions need not disrupt processing even if subjects are highly involved. Second, placing the rhetorical questions at the beginning rather than at the end of the arguments has the advantage of generating interest and curiosity while avoiding the disadvantage of interrupting the train of thought concerning the argument just presented. The results of their study were consistent with this reasoning. The use of introductory rhetorical questions in print enhanced argument elaboration regardless of the personal relevance of the issue (see middle panel of Figure 8-1). Interestingly, despite the similar pattern of attitude data observed under high and low relevance conditions, there was some hint in their thought-listing data that rhetorical questions enhanced elaboration of the message more under low than high relevance conditions. Specifically, an analysis of the total number of thoughts subjects listed about the issue revealed that under low relevance, the rhetorical versions of the message elicited significantly more thoughts than the statement versions, but under high relevance conditions it did not.

Swasy and Munch (1985) conducted a closer replication of our rhetorical study. In this experiment, all subjects heard our strong or weak senior comprehensive exam message in either the statement or the rhetorical form under conditions of high or low personal relevance. The message in this study was presented on audiotape as in our original experiment. Under low relevance, the use of rhetorical questions clearly enhanced elaboration replicating our result and that of Burnkrant and Howard (see bottom left panel of Figure 8-1). However, under high relevance conditions, the attitude measure revealed no evidence for the view that rhetorical questions disrupt processing. In fact, there was little evidence for any effect of rhetorical questions (bottom right panel of Figure 8-1). Nevertheless, on the question asking subjects how distracted they were from thinking about the arguments, an Involvement $\times$ Message form interaction appeared—subjects reported that the rhetorical questions caused more distraction than statements in the high but not the low involvement conditions. Recall that this was the same

pattern that we observed in our initial study (Petty, Cacioppo, & Heesacker, 1981). Swasy and Munch account for the discrepancy between their subjects' own reports that rhetorical questions were distracting under high involvement and the failure to find evidence for this on the attitude measure by "raising concerns about the validity of a self-report distraction measure" (p. 883). An alternative possibility is that the self-reports of distraction are valid, and the attitude results may be attributed to the different procedures used in the studies. Specifically, one potentially important procedural difference between our study and that of Swasy and Munch is that we had subjects complete the thought-listing measure *after* responding to the attitude scales, but Swasy and Munch had subjects respond to the thought-listing measure *before* completing the attitude scales. Subjects may therefore have been motivated and able under high relevance to use the thought-listing period to think about the message and issue prior to completing the attitude measure. Thus, even though the rhetorical questions may have distracted subjects from processing during the message (as revealed in the subjects' self-reports), the thought-listing period provided an opportunity for thought after message exposure. The result of this, of course, would be to attenuate any difference between the statement and the rhetorical versions of the advocacy under high relevance.²

In sum, the research on rhetorical questions has produced consistent support for our view that under conditions of low motivation to process a message, the use of rhetorical questions enhances argument scrutiny. With rhetorical questions, strong arguments become more persuasive but weak arguments become less persuasive (see left column of Figure 8-1). The effect of rhetorical questions under conditions of high relevance is less clear. One possibility is that under conditions of high relevance, the enhancing effect of rhetorical questions on elaboration may be attenuated or even reversed if the message presentation is externally paced. The studies on rhetorical questions provide interesting examples of how an independent variable can have different but explicable effects on elaboration depending on the level of other variables with which it is combined such as personal relevance, modality of message presentation, and postmessage opportunity for thought.

²One unexplained aspect of both replications of our rhetorical study is the failure of the high relevance manipulation to induce greater argument processing. Since this effect has been found several times in independent investigations (e.g., Huddleston, 1985; Leippe & Elkin, in press; Petty & Cacioppo, 1979b, 1984a), it is not clear why it was not obtained in these studies. In any case, the failure of the involvement manipulation to enhance argument processing undoubtedly also contributed to the failure of these studies to find a disrupting effect for rhetorical questions in the high relevance conditions.

Shifting Processing from Objective to Biased

We have seen that some variables, such as the use of rhetorical questions, may enhance elaboration at one level of another factor, but reduce elaboration at a second level of that factor. A second way in which the impact of one variable can depend on the level of another factor is in whether the variable induces relatively objective or relatively biased information processing. For example, we have demonstrated that increasing the degree of personal relevance of a message can enhance a person's motivation to process the message in a relatively objective manner (Petty & Cacioppo, 1979b). However, we have also seen that this processing may become biased if personal relevance is combined with a threat such as that induced by a forewarning of persuasive intent (Petty & Cacioppo, 1979a). In short, the personal relevance may motivate increased processing, but the threat directs this processing in the defense of one's initial position.

Other variables may also interact similarly with personal relevance. For example, we previously noted that all else equal, people would prefer their preexisting attitudes to be correct. Thus, people tend to want to accept proattitudinal messages and reject counterattitudinal ones. In our own research, we have not employed messages that were extremely counter or proattitudinal. If a strongly counterattitudinal message was combined with personal relevance, the net result might well be intensified *unfavorable* processing, whereas if a strongly proattitudinal message was combined with personal relevance, the net result might well be intensified *favorable* processing. In sum, although personal relevance per se may normally enhance a person's motivation to consider the merits of an advocacy in a relatively objective manner, if the message position is either extremely negative or positive, enhanced relevance may still increase processing, but the processing may shift from relatively objective to relatively biased.

Variables that Affect Message Processing and Serve as Peripheral Cues

In the last section we argued that whether or not a particular variable enhances or diminishes processing, or motivates relatively objective or relatively biased processing, may depend on the level of other variables in the persuasion context. Similarly, whether a variable affects information processing or serves as a peripheral cue may depend on the level of other elements in the persuasion context. We discuss this feature of the ELM below.

Multiple Effects of Source Expertise and Attractiveness

In our introductory chapter, we noted that one aspect of persuasion research that has disappointed reviewers of the field is that even variables that were

expected to be quite simple in their effects on attitude change have instead proven to be quite complex. We also noted that perhaps the most dramatic example of this was provided by the conflicting research on features of the message source (Eagly & Himmelfarb, 1974). Postulate 3 of the ELM outlines the several different ways in which source (and other) factors can affect persuasion: they can serve as arguments, they can serve as cues, or they can affect argument processing. In the research that we have reviewed so far, we have focused on how source factors operate when the overall elaboration likelihood is either very high or very low. We have seen that when people are unmotivated and unable to process a message, they rely on simple cues in the persuasion context such as the expertise or attractiveness of the message source, although other cues may be used if they are more salient (see Chapter 6). Importantly, since subjects are either unmotivated or unable to evaluate message arguments, a positive source tends to enhance persuasion and a negative source tends to reduce persuasion regardless of message quality (e.g., see top panels in Figures 6-1, 6-3, and 6-4).

On the other hand, when people are highly motivated and able to process message arguments, strong arguments are more effective than weak ones despite the presence of peripheral cues such as source credibility and attractiveness (e.g., see bottom panels in figures 6-1, 6-3, and 6-4). When motivation and ability to process are high, people are concerned with evaluating the true merits of the advocacy. In order to do this, they will scrutinize all available and inferred information in the immediate persuasion context and attempt to relate it to information stored previously in memory. Importantly, a consideration of source factors may be part of a person's attempt to evaluate issue-relevant information when the elaboration likelihood is high. For example, under some circumstances a source may serve as a persuasive argument (e.g., a physically attractive source may provide persuasive visual testimony as to the effectiveness of a beauty product; Kahle & Homer, 1985; Petty & Cacioppo, 1981b). Additionally, a consideration of source information might help a person in evaluating the true merits of any given argument. For example, consider an expert source (Professor of Education at Princeton) who suggests that tuition be increased at his university. When elaboration likelihood is low (e.g., as a result of low personal relevance or high distraction), this prestigious source might serve as a simple positive cue. When elaboration likelihood is high, however, a subject considering the source and message together might realize that the expert source is biased or has a vested interest in the presentation of some arguments (e.g., an argument to raise tuition to increase faculty salaries). The important point is that when the elaboration likelihood is high, source information does not serve as a simple acceptance or rejection cue, but rather it is considered along with all other available information in the subject's attempt to evaluate the true merits of the arguments and position advocated.

Our conclusions about the impact of source factors under conditions of

high and low elaboration likelihood are only part of the story of how source factors impact on persuasion. As we noted in Chapter 1, we view elaboration likelihood as a continuum anchored at one end by the peripheral route to persuasion, and at the other end by the central route. In all of our research described in the previous sections, we have attempted to create and describe relatively clear instances of central and peripheral routes to persuasion. Thus, for example, in our research on motivation to process, subjects were either highly involved with the topic or very uninvolved. In the high involvement conditions, subjects were confronted with an advocacy that had immediate personal implications for their own graduation (Petty, Cacioppo, & Goldman, 1981) or their finances in attending college (Cacioppo, Petty, & Morris, 1983). In the low involvement conditions, subjects were confronted with a message about a change in university policy that they were *certain* would not affect them, or they were faced with an ad for a product that they were certain would not be available in their local area for a long time (Petty, Cacioppo, & Schumann, 1983). The extreme high and low elaboration likelihood conditions have been quite useful for theory testing purposes and in explicating the two routes to persuasion. However, these conditions represent only part of the elaboration likelihood continuum.

Specifically, many day-to-day persuasion contexts are unlikely to be as high or as low in elaboration likelihood as the conditions we have deliberately created in our initial research. For example, people are sometimes uncertain as to the personal relevance of an issue, or have moderate rather than very high or very low knowledge on a topic. It is important to specify the operation of source (and other) factors when the elaboration likelihood is moderate rather than extreme. We have proposed that under more moderate conditions, people use source (and other) factors to determine how much thinking to do about the message (Petty & Cacioppo, 1981a, 1984c). When a message is on a topic of high personal relevance and people have considerable knowledge on the issue, they know that they want to evaluate the merits of the arguments presented and that they have the ability to do so. Under these conditions, source factors are unimportant in their role as simple cues. When a message clearly is on a topic of very low relevance and/or people have very little issue-relevant knowledge, people know that they either do not want or are not able to think about the message and they seek a simpler way to evaluate the recommendation if a judgment is required. Source cues provide one such simple means of evaluation. However, when the personal consequences or prior knowledge are moderate or unclear, people may not be sure whether the message is worth thinking about. Under these circumstances, characteristics of the message source can help a person decide if the message is worth (or needs) considering. For example, when people have the option of whether or not to think about a message or the proposal is *potentially* relevant, they may reason that it is more worthwhile to think about what an expert has to say

than an inexpert. Why waste precious time thinking about information that is likely to be specious? Or, people may believe that proposals by experts are more likely to be carried out, and are therefore more worthy of thought. A similar case could be made for attractive sources because attractive people are thought to possess a wide variety of positive traits including intelligence (Berscheid & Walster, 1974). On the other hand source expertise may sometimes decrease processing. For example, if people anticipate that others will ask for their opinions, but the issue is one on which they have only moderate knowledge, they may feel that it is relatively risk-free to identify with an expert's opinion but that the arguments of a source whose credibility is suspect require scrutiny. In any case, we have proposed that in many situations people use source cues to determine whether or not a message is worthy of (or needs) scrutiny. Although we have made some suggestions (e.g., Petty & Cacioppo, 1981a, p. 237), future research will be needed to identify the precise conditions under which sources enhance and reduce motivation to process a message.

We have now conducted several studies in which source factors were studied at moderate levels of motivation to process rather than at levels that were very high or very low. We will describe two studies in which we employed the same topics and messages as used in our previous research on personal relevance (see Chapters 4 and 6) to facilitate comparison. Recall that in one study described previously (Petty, Cacioppo, & Goldman, 1981), we told high relevance subjects that a change in academic policy was being considered for next year at their own university (in which case most subjects certainly would be affected personally), and low relevance subjects were told that the policy was being considered for 10 years in the future (in which case current students certainly would not be affected). In our studies on moderate or ambiguous relevance, all subjects were led to believe that a change in policy was being advocated for their university, but they were not told when or if this policy ever would be implemented. Thus, subjects could not be certain whether or not the change in policy would affect them, although there was some potential that if implemented soon, it would. What is the effect of source factors when personal relevance is possible, but uncertain? We have studied both source expertise and source attractiveness under these conditions.

Source Attractiveness Can Affect Message Processing

Method. In an initial study on how attractive sources can affect message processing, 220 undergraduates were randomly assigned to the cells of a 2 (Source attractiveness: high or low) $\times$ 2 (Source age: old or young) $\times$ 2 (Argument quality: strong or weak) between-subjects factorial design (Puckett, Petty, Cacioppo, & Fisher, 1983). The subjects were told that many universities were considering instituting comprehensive exams for seniors and that students in an evening undergraduate continuing education course had written essays on this topic. Each subject was given a folder that

contained a typed essay in favor of the exam proposal along with a card containing a picture and a brief description of the author of the essay. The social attractiveness of the author was manipulated by varying the physical attractiveness of the photograph contained in the folder along with the background information provided: socially attractive authors were described as having better family backgrounds and more prestigious hobbies than the socially unattractive authors. The age of the essay author, was also manipulated (21 vs. 68 years old), but this factor had no effect on persuasion. After subjects examined the folder containing the strong or weak senior comprehensive exam message and the source information, they were asked to rate the essay author and to provide their own opinions about the exam proposal.

Results. Not surprisingly, subjects were more persuaded by the strong than the weak essays. The major result of this study, however, was a significant Argument quality $\times$ Source attractiveness interaction on the measure of subjects' attitudes toward senior comprehensive exams (see Figure 8-2). This interaction was due to the fact that subjects scrutinized the arguments more when they were attributed to the socially attractive sources. In fact, the impact of argument quality on attitudes was significant only for the attractive sources. As depicted in Figure 8-2, this interaction was due to the joint tendencies for attractiveness to enhance agreement with the proposal when the arguments presented were strong, but for attractiveness to reduce agreement when the arguments presented were weak. The latter effect (an attractive source reducing agreement), of course, is opposite to what one normally would expect the effect of attractive sources to be (cf., Chaiken, 1979). Interestingly, a measure of source perceptions indicated that the attractive sources were rated more positively on a number of dimensions (e.g., intelligence, likability, etc.) only when they presented strong arguments. When the attractive sources presented weak arguments, they were derogated relative to the unattractive sources who presented weak argu-

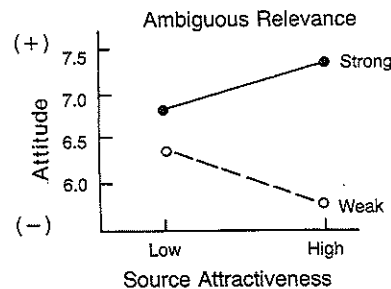


Figure 8-2. Postmessage attitudes as a function of source attractiveness and argument quality under conditions of ambiguous personal relevance (data from Puckett, Petty, Cacioppo, & Fisher, 1983).

ments. This follows directly from the fact that the information presented by the attractive sources was processed more carefully.

Source Expertise Can Affect Message Processing

In a study that was similar conceptually to the Puckett et al. study on source attractiveness, we attempted to determine if source expertise could affect message processing (Heesacker, Petty, & Cacioppo, 1983).

Method. A total of 354 undergraduates were randomly assigned to the cells of a 2 (Source expertise: high or low) $\times$ 2 (Argument quality: strong or weak) $\times$ 2 (Field dependence: field dependent or independent) between-subjects factorial. As in the study on source attractiveness, we again left the degree of personal relevance ambiguous. This time, however, subjects heard rather than read the strong or weak message on senior comprehensive exams. Some subjects were led to believe that the advocacy was based on a six-month study by the Carnegie Commission on Higher Education which was chaired by a Professor of Education at Princeton University (high expertise). Other subjects were led to believe that the advocacy was based on a report of a local high school journalism class chaired by a student in the class (low expertise). The subjects were told that these reports were submitted to the university Chancellor who was seeking recommendations about policy changes to be instituted at their university. Subjects were further told that they were rating the professional broadcast quality of audiotapes that were prepared from written reports by the committees. Finally, subjects in this study were divided into those who were relatively field dependent or independent as assessed by the embedded figures test (Ekstrom, French, & Harmon, 1962). Following exposure to the appropriate message, subjects rated their own attitude on the senior comprehensive exam issue.

Results. Overall, subjects showed more agreement to the strong than the weak message arguments. In addition, however, a three-way interaction was obtained on the measure of subjects' attitudes toward senior comprehensive exams. To understand this interaction, separate 2 $\times$ 2 ANOVAS were computed for field dependent and field independent subjects. The data for field dependent subjects showed the main effect for Argument quality and an Argument quality $\times$ Source expertise interaction ($p < 0.08$; see Figure 8-3). Similar to the effect observed for social attractiveness, the interaction revealed that the arguments were more carefully processed when they were presented by the expert than by the inexperienced source. In fact, simple main effects tests indicated that field dependent subjects differentiated the arguments only when the source was highly credible. As depicted in Figure 8-3, this interaction pattern was due to the joint tendencies for strong arguments to be more persuasive when presented by an expert, but for weak arguments to be less persuasive when presented by an expert, and again the latter effect is opposite to what one normally would expect the effect of

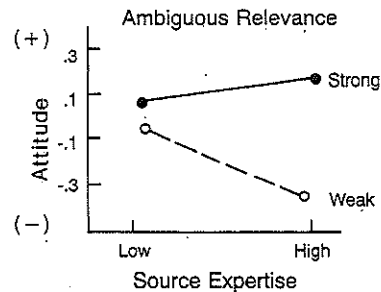


Figure 8-3. Postmessage attitudes for field dependent subjects as a function of source credibility and argument quality under conditions of ambiguous personal relevance (data from Heesacker, Petty, & Cacioppo, 1983).

expertise to be. Field independent subjects, on the other hand, showed only a main effect for argument quality, probably because these subjects are generally more motivated and/or able to extract meaning from stimuli (Witkin, Goodenough, & Oltman, 1979). If field independent subjects generally have a higher elaboration likelihood than field dependent subjects, then they would be more likely to process the message arguments, regardless of the source.

Summary and Discussion

In sum, although the operation of source factors may seem quite simple on the surface, the ELM indicates that their operation, although orderly and predictable, is quite complex. In separate experiments, we have seen that when personal relevance is high, source factors can serve as persuasive arguments or assist in the evaluation of arguments; when personal relevance is low, source factors can serve as simple cues; and when personal relevance is moderate or ambiguous, source factors can affect the extent of message processing (see also Heesacker, 1986; Stoltenberg & Davis, in press). Since these source effects are obtained under different conditions, it is not surprising that a great diversity of results has been observed in the literature. For example, we noted that most research on source expertise has indicated that although expertise effects are rather easy to demonstrate for relatively uninvolved issues, expertise and other simple source effects are less likely for more personally relevant topics (e.g., Chaiken, 1980; Kelman & Hovland, 1953; Petty, Cacioppo, & Goldman, 1981; Rhine & Severance, 1970). Furthermore, the persuasive impact of sources high in expertise has tended to be very short lived (Cook & Flay, 1978). These results are consistent with the ELM in that when an issue is on a topic of low personal relevance, source expertise is postulated to serve as a simple acceptance cue inducing favorable attitudes with little issue-relevant thinking. The consequence is that the attitude formed has little temporal persistence and resistance, and few implications for behavior.

On the other hand, a few studies have shown that expert sources may sometimes produce relatively persisting attitude changes. For example, in a test of his three-process model of persistence (see Chapter 7), Kelman (1958) showed that attitude change induced by an expert source persisted for one to two weeks. In this study, Black college students were presented with a message arguing in favor of maintaining some private all-Black colleges even if the Supreme Court declared segregation unconstitutional. The subjects heard the message just prior to the impending 1954 announcement of the Supreme Court decision on desegregation. Since it was not clear what the Supreme Court would decide about the future of all-Black schools at the time subjects heard the message, whether a new policy would be implemented or not was unclear. These are precisely the conditions under which we have argued that expert sources enhance information processing. If so, then the relative persistence of the attitude changes induced by the expert source in Kelman's study may be attributed to the fact that the expert source enhanced argument elaboration resulting in a well-articulated and persisting attitude.

Other Variables with Multiple Effects

We have now seen how some source variables can serve as arguments in some contexts, cues in other contexts, and affect argument processing in still other situations. This general principle, which is stated explicitly in ELM Postulate 3 (see Table 1-1), was applied mostly to separate variables elsewhere in this volume. However, it should now be clear that any one variable can serve in all of these roles. A few more examples should help to elucidate how one variable can serve in multiple roles depending on the specific features of the persuasion context.

Multiple Sources

First, consider the impact of the number of other people who endorse a particular attitudinal position. As we noted in Chapter 4, traditional analyses of the number of message sources have assumed that the more people who are perceived to advocate a position, the more conformity pressure that is induced, and the more agreement that results (e.g., Asch, 1951). Whether or not the sources presented arguments in support of their positions was unimportant. People agreed with multiple sources out of a desire to be correct (Festinger, 1954) or at least to appear correct (Jellison & Arkin, 1977). However, more recent analyses have argued that multiple sources may enhance information-processing activity (e.g., Burnstein & Sentis, 1981; Harkins & Petty, 1983). For example, recall from Chapter 4 that Harkins and Petty (1981a) found that having three sources present strong arguments enhanced persuasion but that having three sources present weak arguments reduced persuasion over that achieved when the same information was presented by one source (see Figure 4-5).

Importantly, the Harkins and Petty (1981a) study was conducted under conditions of ambiguous personal relevance and the results obtained were similar to the effects we reported in this chapter for source attractiveness and expertise under similar conditions (see Figures 8-2 and 8-3). According to the ELM, however, what would be expected to happen if message relevance was very high or very low? When the elaboration likelihood is very low (such as when personal relevance or prior knowledge are very low, distraction is very high, etc.), people will be unmotivated and/or unable to engage in the cognitive work necessary to evaluate the issue-relevant arguments. Under these conditions, people may use the number of people who support the issue as a simple cue as to the worth of the proposal. Therefore, judgments would be based primarily on this cue—the more people who supported the proposal, the more agreement that would result regardless of whether the arguments presented were strong or weak. When the elaboration likelihood is very high, however, message recipients will undertake a deliberate assessment of the message arguments and the number of endorsers will have little value as a simple acceptance cue. In short, the ELM holds that the number of people supporting a message may serve as a simple conformity cue under some conditions (low elaboration likelihood), act to enhance processing in other conditions (moderate elaboration likelihood), and have no power as a simple cue in still other conditions (high elaboration likelihood). These different predicted effects are depicted in the top three panels of Figure 2-3. No experiment to date, however, has examined the impact of the number of sources across the full elaboration likelihood continuum.

Message Variables

Factors associated with the message source, of course, are not the only variables that can both serve as cues and affect message processing. Message variables can likewise serve in both roles. We have already discussed in Chapter 6 how the number of arguments could serve as a simple cue when personal relevance was low, but affect information processing when personal relevance was high (Petty & Cacioppo, 1984a). Other message variables may operate similarly.

For example, earlier in this chapter we argued that the use of rhetorical questions could increase elaboration when personal relevance was low, but reduce elaboration when personal relevance was high (for audio messages). Can rhetorical questions also serve as simple cues? Since some research suggests that people may make various inferences about the message source depending upon the kind of language used (Ryan & Giles, 1982), rhetorical questions have the *potential* to serve as cues. Research has suggested that sometimes the use of rhetorical questions makes the speaker appear more polite (a positive cue; e.g., Bates, 1976), but sometimes the use of rhetorical questions makes the speaker appear less confident (a negative cue; e.g., Newcombe & Arnkoff, 1979). If a communication were on a very esoteric or

unfamiliar topic for which the audience had few or no preexisting message-relevant beliefs, it would be unlikely that the use of rhetorical questions could affect message elaboration. For such issues, to the extent that some evaluation was required, rhetorical questions might serve as simple cues. To the extent that the rhetorical questions made the speaker or the arguments appear more favorable, more agreement would result (cf., Zillmann, 1972), but to the extent that the rhetorical questions made the speaker or arguments appear less favorable, less agreement would result.

Recipient and Context Factors

In addition to source and message variables, recipient and context variables may also serve in multiple roles. For example, in Chapter 3 we saw how the physical posture of a message recipient (reclining or standing) could affect the extent of elaboration under moderate involvement conditions (see Figure 3-7; Petty, Wells, Heesacker, Brock, & Cacioppo, 1983). If subjects were presented with a message they were unable to elaborate, then posture or other factors related to comfort during message exposure might serve as simple positive or negative affective cues (e.g., Griffitt & Veitch, 1971).

In Chapter 6 we saw how false physiological feedback could serve as the basis of a simple inference about the desirability of a picture when the personal consequences of doing so were low (Taylor, 1975). However, in other situations, physiological feedback may cause a person to ruminate about *why* a picture (or other stimulus) has affected bodily processes. This may involve extensive processing of the attributes of the stimulus and lead to attitude changes that are induced by the central rather than by the peripheral route (cf., Liebhart, 1979). In any given situation, depending on the nature of the stimulus employed (e.g., involving or not), the type of bodily response detected (felt or not), the amount of time provided to think, and other factors, the change induced may either be the result of a simple inference or may result from more extensive processing of the stimulus (Petty & Cacioppo, 1983b).

The Role of Affect in the ELM

Although persuasion researchers over the past few decades have focused on the cognitive foundations of attitudes, investigators are beginning to show renewed interest in the affective bases of attitudes and other judgments (e.g., Cacioppo & Petty, 1981a; Clark & Fiske, 1982; Zajonc & Markus, 1982). Importantly, the ELM holds open the same possibilities for affect that it does for the other source, message, recipient, and context variables that we have already discussed. For example, we noted in Chapter 1 that for some people or in some situations, a determination of the central merits of an attitude object might entail an analysis of one's *feelings* rather than one's beliefs or behaviors. Thus, the affective state (e.g., love) induced by an attitude object (e.g., a person) might serve as a persuasive argument for or against the merits of the object when the elaboration likelihood is high (e.g., when the person is a potential spouse). This process is most likely when the

affect is perceived to be directly relevant to the central merits of the object or issue under consideration (e.g., the salivating joy one feels when considering the merits of consuming a sizzling steak when famished, or the disgust one feels after realizing that the steak is rancid). In short, according to the ELM, relevant affective states should serve as persuasive arguments or help in assessing the cogency of arguments when the elaboration likelihood is high (e.g., fear may contribute to a person's assessment of the danger inherent in not following a specific recommendation; cf. Rogers, 1983). When the elaboration likelihood is high but *irrelevant* affective states are induced, it is likely that the affect will bias issue-relevant thinking by making affectively consonant thoughts and ideas more accessible in memory (Bower, 1981; Clark & Isen, 1982). As we noted in Chapter 5, this process may explain the increased negative thinking that is observed when strong arguments are repeated an excessive number of times (cf., Cacioppo & Petty, 1985).

In contrast to the postulated impact of affective states when the elaboration likelihood is high, when people are either relatively unmotivated or unable to engage in the cognitive work necessary to evaluate the central merits of an attitude object or issue, then affect, whether relevant or irrelevant, should serve as a simple peripheral cue. As a cue, affect should enhance attitudes when it is pleasant, but have a negative effect when it is unpleasant. In Chapter 6 we reviewed research that was consistent with this principle. For example, recall that Gorn (1982) found that pleasant music had a favorable impact on attitudes toward a consumer product but only under conditions of product relevance. Similarly, Srull (1983b) found that mood had an impact on attitudes for people who were low in issue-relevant knowledge, but did not influence attitudes when knowledge was high.

Finally, we note that when people are uncertain as to whether or not they wish to scrutinize a persuasive communication (moderate elaboration likelihood), one's affective state may determine whether or not the communication is processed. Specifically, people may be more willing to undertake the cognitive effort necessary to process a message if they are feeling good rather than bad. The basis for this reasoning stems from previous research suggesting that disphoric states tend to dampen one's level of activity including the tendency to engage in effortful cognitive endeavors (e.g., see Castillo, 1985). Interestingly, the Petty, Wells, Heesacker, Brock, and Cacioppo (1983) study on recipient posture (graphed in Figure 3-7) can be interpreted in this context. To the extent that standing was less pleasant than lying during the message, this relative discomfort might have contributed to the diminished processing observed when people were placed in this posture. In sum, although research on the role of affect in persuasion is sparse at present, the ELM provides a guiding framework for analyzing its role in attitude change.³

³Our analysis of affect presumes a *moderate* level of this construct (i.e., not so low as to be imperceptible, but not so high as to be alarming or disruptive of all processing).

Summary

It is important to note that even though the ELM holds open the possibility that variables can affect agreement either by having an impact on information processing or by serving as simple cues, the ELM specifies, in a general manner at least, the conditions under which each process is likely to operate. Thus, a whole list of source (e.g., credibility, attractiveness, number of sources), message (e.g., number of arguments, use of rhetorical devices, discrepancy), audience (e.g., affective states, presence of hecklers, false physiological feedback), and other variables may affect attitudes by modifying information processing under certain conditions (e.g., moderate to high personal relevance), but affect attitudes by serving as simple cues in other contexts (e.g., low prior knowledge).

Processing When No Message or Both Sides Are Presented

Throughout most of this volume we have focused on the typical persuasion situation in which various arguments on only one side of an issue are presented to a recipient. Other persuasion contexts occasionally arise, however, and in this final section we briefly address the implications of the ELM for two of them.

First, in some situations a position is advocated, but no arguments are presented for the person to evaluate. Still, according to the ELM, motivation and ability to think will be important determinants of the route to persuasion. However, when no arguments are presented, thinking cannot be guided by the message arguments. Instead, thinking can be guided either by the person's initial attitude, or by the topic of the advocacy itself. In the former case, people will engage in a biased scanning of arguments in memory supporting their initial positions, and attitude polarization may result (see Tesser, 1978). In the latter case, people will search memory in a more objective manner for information and arguments that are consistent with the advocated position. To the extent that supportive information and arguments are available, change in the direction of the advocacy may occur (see Burnstein & Vinokur, 1977; King & Janis, 1956). Importantly, in each of these cases, the attitudes formed are the product of issue-relevant thinking and will have the characteristics of other attitudes induced via the central route. Of course, if no arguments are presented, and people lack the requisite motivation or ability to think about the issue, then peripheral cues associated with the advocacy may determine the extent and direction of influence.

A second atypical situation occurs when people are presented with arguments on both sides of an issue. Again, motivation and ability to think about the issue will determine the route to persuasion. If motivation and/or ability are low, but a judgment is required, people will examine both sides in a cursory manner and will likely favor the position associated with the more favorable cues. For example, the side with the greater number of

arguments, or the side endorsed by the more prestigious source will be selected. However, if people have the necessary motivation and ability to evaluate the issue in a relatively objective manner, then the side presenting the highest quality arguments will be favored. If both sides present equally compelling positions, people may be motivated to resolve the ambiguity by using the available information to bolster their initial positions. Also, if one's previous knowledge on a issue is biased in favor of one's initial position, as will often be the case, then people will likely have greater ability to cognitively bolster their side but counterargue the opposing side. This process can lead to attitude polarization (e.g., Lord, Ross, & Lepper, 1979; see Chapter 5).

Retrospective

In this chapter we have examined and addressed some of the complications that arise in the Elaboration Likelihood Model. Among the complications are that variables can have one effect on elaboration when they are paired with some variables, but have different, even opposite, effects on elaboration when they are paired with other variables. Thus, some variables enhance elaboration at one level of a factor, but disrupt elaboration at another level of the same factor. Additionally, some variables affect information processing in a relatively objective manner when combined with some factors, but affect information processing in a more biased fashion when combined with other factors. Finally, we noted that the same factor can serve in any one of the three major roles postulated for variables by the ELM. That is, any one factor (e.g., source attractiveness) can serve as a persuasive argument in one context, act as a peripheral cue in another, and affect message elaboration in still a third situation. Fortunately, the ELM provides guidance as to when each of these processes is likely. In the final chapter, we offer some concluding remarks about the ELM.

Chapter 9

Epilogue

Introduction

We began this monograph with questions about the likely effectiveness of televised speeches given on New Year's Day by the American president and the Soviet premier to national audiences in each other's countries. The research and theory presented in this volume have indicated that communications such as these can produce persuasion via two fundamentally different routes. One route is based on the thoughtful (though sometimes biased) consideration of arguments perceived central to the merits of the issue under consideration, whereas the other is based on affective associations or simple inferences tied to peripheral cues in the persuasion context. When variables in the persuasion situation render the elaboration likelihood high, the first kind of persuasion occurs (central route). When variables in the persuasion situation render the elaboration likelihood low, the second kind of persuasion occurs (peripheral route). Importantly, there are different consequences of the two routes to persuasion. Attitude changes via the central route appear to be more persistent, resistant, and predictive of behavior than changes induced via the peripheral route (see Figure 1-1).

In outlining the antecedents and consequences of the two routes to persuasion, the Elaboration Likelihood Model provides a framework from which to analyze the speeches of the U.S. and Soviet leaders. For example, a fundamental question is: How much thinking were the audiences likely to be doing about the communications? First, recall that the American president appeared during a news program in the Soviet Union, whereas the Russian premier appeared during televised coverage of a holiday parade in the United States. Thus, the elaboration likelihood was probably higher for the former than the latter speech. Because of this, the American leader may have been more likely to be judged on the quality of his arguments, whereas peripheral cues may have had a greater impact on attitudes toward the